V&V Summary Report
L2 ASCDS Version : 10.12.6

Observation 631 - L2 Version 5
Chandra X-Ray Center

L2 Processing Date : Jun 25 2024

See axaff00631N002_VV001_vvref2.pdf for the full report

V&V Scientist	Jen Lauer
V&V Date (YYYY-MM-DD)	2024.06.28
V&V Edition	1
V&V Disposition and Status	OK
V&V Charge Time	37.126

Comments

```
Slot 3 was not utilized in this observation.
===
Charge time for this ObsId remains at previous value of 37.126
ks,although with the current processing the charge time would have been
37.165 ksec.
===
The focal plane temperature during the interval 59228327.36 -
59265488.96 (MET s) of this observation was warmer than the upper limit
for optimum calibration of the ACIS gain and spectral resolution (i.e.,
-111.0 C for ACIS-S).
The Chandra calibration team calibrates the ACIS gain and spectral
resolution using data from the external calibration source (ECS). ECS
data show that the frontside-illuminated (FI) CCDs are more temperature
sensitive than the backside-illuminated (BI) CCDs.
A summary of the current calibration status of the ACIS gain and
spectral resolution can be found at:

http://asc.harvard.edu/cal/Acis/Cal_prods/Gain_and_Spectral_Resolution/A
CIS_response_summary.html
```

The main points are:
1) The gain on BI chips remains within 0.3% (i.e., the systematic uncertainty in the ACIS gain quoted on the Chandra Calibration Status Summary web page) at all measured temperatures.
2) The gain on FI chips remains within 0.3% below row 600 at all measured temperatures.
3) The gain on FI chips above row 600 can be underestimated by as much as 1% for focal plane temperatures exceeding -116 C.
4) The spectral resolution (i.e., FWHM) on BI chips is insensitive to the focal plane temperature.
5) Warmer focal plane temperatures increase the FWHM on FI chips by up to 30 eV near row 512 and by up to 70 eV near the top of the chips.

In summary, the user should be cautious in the spectral analysis of high S/N emission lines detected on the top half of FI chips in this observation. Default processing with the current version of the CALDB will underestimate photon energies by up to 1% and broaden emission lines by up to 70 eV.

seq_num	200056	Sequence number
obs_id	631	Observation id
title	SHOCKED STELLAR WINDS IN PLANETARY NEBULAE	Proposal title
observer	Prof. You-Hua Chu	Principal investigator
object	NGC 7293	Source name
dtycycle	0	
cycle	P	events from which exps? Prim/Second/Both
ra_targ	337.410417	[deg] Observer's specified target RA
dec_targ	-20.838333	[deg] Observer's specified target Dec
ra_nom	337.40998486989	[deg] Nominal RA
dec_nom	-20.843581546606	[deg] Nominal Dec
roll_nom	296.4914650502	[deg] Nominal Roll
revision	5	Processing version of data
ontime	37164.595832668	[s] Sum of GTIs
livetime	36693.995342402	[s] Livetime
ontime5	37164.554792665	[s] Sum of GTIs
ontime6	37164.513752669	[s] Sum of GTIs
ontime7	37164.595832668	[s] Sum of GTIs
ontime8	37161.231762446	[s] Sum of GTIs
ontime9	37161.190692529	[s] Sum of GTIs
l2events	424817	Number of level 2 events

